6.6.3

a) The training RSS will steadily decrease because the total summation of the linear regression's coefficients will grow from 0, allowing for a better fit for the training data. Otherwise, the model is based off the null hypothesis/value which has the highest training error at $s=0$.

b) The test RSS will decrease initially, and then eventually increase in a U shape. Initially, the co-efficients are too weak when s is near 0, leading to a constrained model and high test error. Next, as s increases, the model becomes more flexible, detecting underlying patterns, decreasing test error. However eventually, the coefficients and the model become too overfit as s increases more, causing the test RSS to increase again.

c) Variance steadily decreases. When $s=0$ or is near 0, the model has zero-to-little variance since all coefficients are restricted to being insanely small, or just 0. With s increasing, the coefficients grow too, allowing for a more flexible model, and for coefficients to be more sensitive to fluctuations in the training data.

d) Squared bias steadily decreases. As s increases, the model is allowed to fit more complex relationships, otherwise at $s=0$, the model is restricted to a flat constant line with high bias.

e) Irreducible error remains constant because these are errors the model can't explain nor understand, so it's completely independent from the model and its flexibility.

6.6.5

a) So the standard ridge regression objective function is:

$$\min_{\beta_0 \to \beta_0} \underbrace{\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2}_{\text{main part}} + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\text{penalty}}$$

and since $\beta_0 = 0$, $n=2$, and $p=2$, the main part, when expanded:

$$(y_1 - (\beta_1 x_{11} + \beta_2 x_{12}))^2 + (y_2 - (\beta_1 x_{21} + \beta_2 x_{22}))^2$$

Since $x_{11} = x_{12} = x_a$ (arbitrarily let's say x_a)

$x_{21} = x_{22} = x_b$

and, because $y_1 + y_2 = 0$, $y_1 = -y_2 = y$ and thus:

$$(y - x_a(\beta_1 + \beta_2))^2 + (-y - x_b(\beta_1 + \beta_2))^2$$

Since $x_{11} = -x_{21}$ and $x_{12} = -x_{22}$, $x_b = -x_a$, and:

$$(y - x_a(\beta_1 + \beta_2))^2 + \underbrace{(-y + x_a(\beta_1 + \beta_2))^2}_{\text{equal to } (y - x_a(\beta_1 + \beta_2))^2 \text{ since } (x)^2 = (-x)^2}$$

so our specific function becomes:

$$\min_{\beta_0 = 0, \beta_1, \beta_2} \left[2(y - x_a(\beta_1 + \beta_2))^2 + \lambda(\beta_1^2 + \beta_2^2) \right]$$

6.6.5 CON'T.

b) Our objective function only depends on β_1 and β_2 through their sum $\beta_1+\beta_2$ and squared sum $\beta_1^2+\beta_2^2$. Swapping one for the other won't lead to a different result per the commutative property of addition, meaning both terms β_1 and β_2 are symmetric with respect to one another, implying that this minimizer is true: $\hat{\beta}_1 = \hat{\beta}_2$.

c) The only thing different with LASSO's objective is that it penalizes the absolute value of the coefficients — not their squares, so we easily get:

$$\min_{\beta_0=0,\beta_1,\beta_2} \left[2(y-(\beta_1+\beta_2))^2 + \lambda(|\beta_1|+|\beta_2|)\right]$$

d) Because we take the absolute values of the coefficients, they could be considered the same whether it is their positive or negative counterpart. Moreover, because we only care about the result of $\beta_1+\beta_2$, any combination of β_1 and β_2 that lead to the same optimal sum work, resulting in multiple possible optimal solutions.

8.4.5

Per the standard majority vote approach, we would do P(Red $|$ X > 0.5), otherwise green.

so with the samples: 0.1 (G), 0.15 (G), 0.2 (G), 0.2 (G), 0.55 (R), 0.6 (R), 0.6 (R), 0.65 (R), 0.7 (R), 0.75 (R)

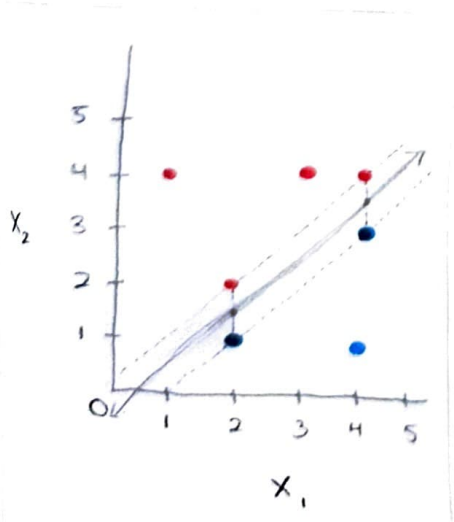
we get 4 Greens and 6 Reds, so Red wins.

Per the average voting approach:

avg val $= \frac{\text{sum_vals}}{10} = 0.45$. Since 0.45 falls under the 0.5 threshold, this approach classifies the result as Green.

9.7.3

a+b)



The two closest red points lie on $X_1 = X_2$ and the closest blue points on $X_2 = X_1 - 1$ so the boundary, halfway between, is:

$$X_2 = X_1 - 0.5$$

or

$$0.5 - X_1 + X_2 = 0$$

c) Classify to Red if $0.5 - X_1 + X_2 > 0$ and classify to blue otherwise.

$\beta_0 = 0.5$

$\beta_1 = -1$

$\beta_2 = 1$

9.7.3 CON'T.

d) Since red points: $(2,2)$ and $(4,4)$

blue points: $(2,1)$ and $(4,3)$, perpendicular distance is:

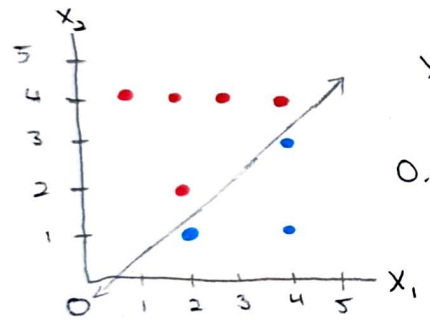
$$\text{distance} = \frac{|1(2) - 1(2) - 0.5|}{\sqrt{1^2 + (-1)^2}} = \frac{0.5}{\sqrt{2}} = \frac{\sqrt{2}}{4} \approx 0.354$$

e) blue SV: $(2,1)$ & $(4,3)$

red SV: $(2,2)$ & $(4,4)$

f) The 7th point is notably far away from the blue support vectors. Since the hyperplane is only determined by the relevant SV's, slightly moving that 7th observation would not affect the hyperplane, given that it moving "slightly" means it goes nowhere near the margin.

g).

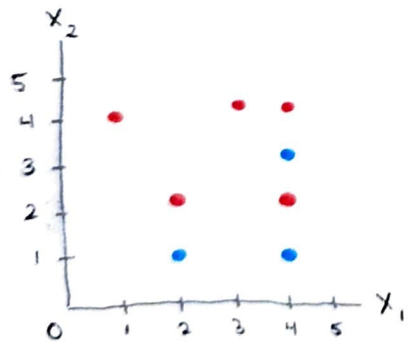


$$X_2 = X_1 - 0.75$$

or

$$0.75 - X_1 + X_2 = 0$$

h)



New red point at $(4,2)$ makes this inseperable! (linearly)